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PROFILE

An initiative-taking and technically accomplished 8th-grade middle school student with an exceptional aptitude for computer science and software engineering — self-taught in Python, JavaScript, Electron, HTML, and Flask, and the sole architect and builder of a production-ready AI platform serving real users, launched at age 12. The Wizard Ecosystem has since grown into fourteen fully integrated, live applications — an AI assistant, desktop browser, search engine, email client, calendar, real-time chat, AI-powered note-taking application, computer automation system, developer API, a personal finance manager, extreme-weather/wildfire safety tools, a desert wildlife companion, and a full AI-powered code editor. Beyond software development, a First-Degree Black Belt in Taekwondo (earned April 2026) reflects the same discipline, resilience, and perseverance applied to every technical endeavor. Seeking internship, scholarship, or collaborative opportunities to leverage these skills toward meaningful, real-world impact.

FEATURED PROJECT



Wizard Ecosystem — Complete AI Platform

Mar 2026 – Present

Live: wizardecosystem.dev

Tech Stack: Python (Flask), JavaScript, Electron, SQLite, Groq API (Llama 4 Scout), HTML/CSS, Node.js

Built a production-ready, fully integrated AI ecosystem from scratch — fourteen applications in one live platform, deployed on PythonAnywhere (backend), GitHub Pages (frontend), and Electron desktop installers for Windows, macOS, and Linux.



Wizard AI (v15.1.0) — Multi-Agent Assistant

- Seven built-in AI personalities (JARVIS, ORACLE, Nerd, Fun, Sarcastic, Fast, Normal); custom personality creator with emoji, system prompt, community sharing, and like/usage tracking stored in SQLite.
- RAG pipeline: ingests PDF, DOCX, XLSX, XLS, TXT, CSV, JSON, HTML, CSS, JS, PY, MD — extracts full text (PyPDF2, python-docx, openpyxl), splits into 1,500-character chunks with 100-word overlap, keyword extraction, intersection/union similarity scoring, top-5 chunks injected into Llama 4 Scout prompts.
- Multi-agent orchestration: 9 agent types (Researcher, Writer, Coder, Designer, Data Analyst, Reviewer, Summarizer, Translator, Optimizer); Chain Studio with pre-built workflows (Research & Write, Code & Review, Data Analysis, Translate & Summarize) and a custom chain builder; async sequential execution via SSE streaming.
- WizProfile — rich structured user profile system replacing simple key-value memory: 50+ fields across Basic Info (name, birthday, bio, avatar), Professional (occupation, company, skills, experience), Interests (hobbies, topics, learning goals), AI Preferences (preferred mode, response style, formality level, emoji preference), and Goals & Tasks (goals with deadlines, reminders); profile completeness meter (0–100%) with auto-sync across all ecosystem applications.
- AI personalization via WizProfile: profile context injected into every prompt so the assistant addresses users by preferred name, matches desired formality and response style, references skills and interests, and follows custom per-user instructions and greetings.

- AI auto-learning extracts profile information naturally from conversation; import capabilities from LinkedIn URL, GitHub username, or resume upload (PDF/DOCX); profile stored in dedicated `wiz_profile` table (FK to User) with JSON columns for structured fields including `social_links`, `education`, `skills`, `interests`, `goals`, and `favorites`.
- Web search with multi-engine fallback: DuckDuckGo Lite → DuckDuckGo HTML → Brave Search; DDG redirect removal to expose real destination URLs.
- Image generation with multi-service fallback: Pollinations.ai → Hugging Face (Stable Diffusion) → Lexica → Unsplash; square/landscape/portrait sizes; base64 for direct display and download.
- Secure Python sandbox: stdout/stderr capture, blocked dangerous imports (`os`, `subprocess`, `importlib`, `eval`, `exec`), safe globals only, timeout and memory limits.
- Streaming responses via Server-Sent Events; chat history persisted in JSON per user/chat; session security with HTTP-only, `SameSite=None`, Secure cookies.



Wizard Notes — AI-Powered Note-Taking Application

- Full-featured, AI-powered note-taking application integrated directly into the Wizard Ecosystem, accessible via a responsive web interface with a dark purple/gold aesthetic consistent with the broader platform.
- Core note management: rich text notes with titles and content created via modal; click-to-edit with auto-save on keystroke; delete, favorite (star), and rename via dedicated modal; Recent Notes view automatically surfaces the 10 most recently edited entries.
- Organization and navigation: custom tag creation with tag-based filtering; folder views (All Notes, Favorites, Recent); full-text search (in development); left-side panel for notes list, tags, and folder navigation alongside a clean, distraction-free editor.
- AI-powered editing suite: AI Edit for direct note transformation (e.g. "Make this more professional"); Grammar Fix for instant spelling and grammar correction; Summarize to generate a 2–3 sentence synopsis; Format to apply bullet points and structured headings; Translate to convert note content into any target language; Expand to elaborate and enrich brief drafts — all accessible via an always-visible AI Panel.
- Sharing and collaboration: share notes with any Wizard Ecosystem user by email address; permission levels include read-only (Can View) and full access (Can Edit); Shared Inbox for receiving notes from others; real-time synchronization ensures changes propagate immediately across all collaborators.
- Technical architecture: frontend at `/notes/index.html`; RESTful backend endpoints (GET/POST `/api/notes`, PUT/DELETE `/api/notes/{id}`, POST `/api/notes/{id}/favorite`, POST `/api/notes/{id}/share`, GET `/api/notes/shared`); note data stored as JSON within the SQLite `user.notes_data` column; authentication via the same session-based system as Wizard AI.



Wizard Autopilot — Full Computer Automation

- Natural-language command parser (regex-based, no external API): resolves commands such as "open amazon", "search for gaming laptop on Walmart", and "create a modern blog about AI" entirely locally.
- Full OS automation from Electron's main process — no Python server dependency: Windows via PowerShell with `System.Windows.Forms` (`Cursor.Position`, `SendKeys`, `System.Drawing`); macOS via `osascript`/System Events for mouse and keyboard, screenshot for screenshots; Linux via `xdotool` for mouse/keyboard and `import` for screenshots.
- Full capability set: cursor positioning, left/right/double-click, scroll, text typing, individual key presses (Enter/Tab/Esc/Backspace/Arrows), shortcuts (Ctrl+C, Ctrl+V, Alt+Tab), URL and embedded browser control, Amazon/Walmart/Craigslist/Google search, AI-generated HTML/CSS file creation, directory listing, file reading, screenshot capture, and system information retrieval (OS, CPU, RAM, username, hostname).
- Chat-style UI with command examples sidebar, real-time status, file preview, screenshot display, system info cards, and typing indicator.



Wizard Browser (v10.0.0) — Full Desktop Web Browser

- Multi-tab management: create, close, and switch tabs; favicon and title updates; Ctrl+T / Ctrl+W shortcuts; tab persistence across sessions.

- Navigation: back/forward with array-index history, reload, URL bar with DuckDuckGo search suggestions (debounced), smart URL vs. search-query detection.
- Workspaces: Default / Work / School / Media (customizable); each workspace holds independent tabs and active tab state; creation modal with name and emoji picker (20+ options); persisted to localStorage.
- Password manager: auto-detects credentials via injected DOM listeners on login form submission; domain normalization; CSV import/export (Brave/Chrome compatible); one-shortcut autofill (Ctrl+Shift+F).
- Ad blocker (doubleclick.net, googleadservices.com, googlesyndication.com, facebook.com/tr); bookmarks panel; settings panel with Purple/Light theme, search engine selector, private mode, and data management.



Wizard Search — AI-Powered Search Engine

- Pipeline: query → DuckDuckGo Lite fetch → regex table parser → DDG redirect removal (uddg param) → JSON results.
- Fallback chain: DDG Lite → DDG HTML → Brave Search → raw Google/DDG links.
- AI summaries: Groq Llama 4 Scout generates a 2-3 sentence topic overview per query, educating the user beyond simple result listings.
- Real-time search suggestions via DuckDuckGo autocomplete API, debounced at 300 ms.



Wizard Mail — Full Email Client

- IMAP/SMTP support for Gmail (App Password), Outlook, Yahoo, and custom domains; auto-detects servers from email domain; persistent connection via pickle file.
- Folders: INBOX, Sent, Spam, Trash, Drafts, Starred; Gmail folder mapping ([Gmail]/Sent Mail, [Gmail]/Spam, etc.).
- Operations: fetch last 50 emails newest-first; parse plain text and HTML bodies including inline CID images; reply with proper threading headers (In-Reply-To, References, Message-ID); star/unstar; delete (\Deleted + EXPUNGE).
- Email aliases: instant @wizardecosystem.dev address provisioning via Cloudflare Email Routing; Gmail “Send As” setup instructions provided.



Wizard Calendar — Event Management

- Month and week views; event creation with title, start/end datetime, color coding (violet/blue/green/red/gold), description, and location; mini-calendar for quick navigation.
- Events stored per user in SQLite (CalendarEvent model); JavaScript date mathematics for grid rendering including first-day-of-month, days-in-month, and event overlap detection.



Wizard Chat — Real-Time Messaging

- Real-time message delivery via 2-second polling (tab-focus aware); typing indicators (3-second timeout, cleared on focus loss); read receipts with unread counts in sidebar.
- SQLite models: Conversation (user1_id, user2_id, timestamps) and Message (conversation_id, sender_id, content, is_read, read_at, created_at); user search by email or name.



Developer Hub & API

- OpenAI-compatible REST API: POST /v1/chat/completions, GET /v1/models, POST /api/keys/create, GET /api/keys, POST /api/keys/<id>/revoke, GET /api/stats.
- Bearer token authentication (wiz_... prefix); keys stored with bcrypt hashing; per-key request counting and last-used tracking.
- Python SDK published to PyPI as wizard-ai-pro; Admin Dashboard with user management, system logs, server status, backup, cache clearing, maintenance mode, broadcast messaging, and JSON statistics export.



Wizard Finance — AI-Powered Personal Finance Manager

- Dashboard with income/expense/net savings summary cards, month-over-month change indicators, top spending category, and a Chart.js doughnut chart of spending by category.
- Transaction management: add income/expense entries with amount, category, merchant, and date; live filters by type and category; paginated transaction history with infinite “Load More.”
- AI receipt scanner: in-browser camera capture with a capture-guide overlay; captured image sent to the backend for AI extraction of merchant, amount, category, and date, auto-filling the transaction form.
- Budgeting: per-category monthly budgets with configurable alert thresholds and an automatic budget-alert banner when spending approaches the limit.
- Savings goals: create goals with target amount, category, and optional target date; progress bar with percentage tracking and one-click contributions toward each goal.
- AI Insights tab: natural-language financial summary and monthly income/expense/net breakdown generated from transaction history.



Wizard Heat — Extreme Weather & Hydration Assistant (Arizona)

- Real-time heat-risk dashboard: temperature, “feels like,” humidity, wind, and UV index, with a color-coded risk badge that also tints an ambient horizon-glow UI element to match current risk level.
- Hydration tracker: circular progress ring toward a daily ounce goal with quick-add buttons (+8/+16/+24 oz), persisted per user via the backend API.
- 5-day forecast with per-day high/low and risk-level tags; Fahrenheit/Celsius toggle applied across the dashboard and forecast.
- Cooling-center locator and an activity-safety advisor (hiking, running, walking, biking, gardening, sports, pool, picnic) returning a safety status, advice, and recommended water intake.
- Client-side-only tools: configurable hydration-reminder notifications, a sunscreen-reapplication countdown timer, and a one-click “copy current conditions” summary.



FireWatch AI — Real-Time Wildfire Detection

- Live status dashboard aggregating NASA FIRMS, InciWeb, and NWS Alerts, with a color-coded fire-risk meter (Low/Moderate/High/Extreme) driven by temperature, humidity, wind speed, and active alert count.
- Active fire tracking: severity-coded list with distance, containment status, and acreage; ASCII fire-map preview that repositions a fire icon based on the nearest active fire’s distance.
- Automatic critical-alert banner surfaced above the dashboard when a fetched alert is flagged critical; recent-alerts feed with timestamps and severity badges.
- AI-generated evacuation-route suggestions and a smoke forecast (per-time-block severity bars) alongside live wind direction and speed.
- Location search with geocoding and one-click “My Location” geolocation, including client-side reverse-geocoding to resolve coordinates into a readable place name; auto-refreshes every 2 minutes.



Wizard Nature — Desert Survival & Wildlife Companion

- Cactus Tracker: spot-and-log interface for six Sonoran Desert cactus species (saguaro, barrel, prickly pear, cholla, organ pipe, senita) with persisted spotted/unspotted state.
- AI photo identification suite built on vision AI: Snake ID (rattlesnake vs. harmless), Critter ID (lizards, birds, wildlife), and Plant ID (cacti, wildflowers, desert plants), via drag-and-drop or tap-to-upload with markdown-rendered AI analysis.
- Sunset Forecast: nightly 0–10 quality rating from cloud cover, humidity, and air-quality inputs with a plain-language recommendation, auto-refreshed every 5 minutes.
- Wildflower Alert: post-rain bloom-season detector with best-viewing-spot suggestions and an opt-in bloom notification.
- Camping Guide surfacing rated Arizona campsites with amenities, season, and reservation info; ambient UI glow shifts color with real local time of day (dawn/midday/dusk/night).



Wizard Code — AI-Powered Desktop Code Editor

- Full Electron-based IDE built on the Monaco editor: multi-tab editing, side-by-side split view, drag-and-drop file explorer with inline create/rename/delete, integrated terminal with live stdin, and a VS Code-style command palette.
- Wizard AI chat panel with quick actions (Explain, Fix, Generate, Comment), streaming responses, ghost-text inline AI code completions, and hover-triggered AI error explanations with one-click “Fix with Wizard AI.”
- Agent Mode: autonomous multi-file AI agent with function-calling tools (list_files, read_file, write_file) that stages changes without touching disk; changes reviewed and accepted or rejected per-file or in bulk via a Monaco diff viewer.
- Real background diagnostics: live Monaco language-service validation for JS/TS/JSON/CSS/HTML, plus debounced compiler/linter checks (py_compile, javac, gcc, rustc, go vet, bash -n, php -l, ruby -c) parsed into accurate error/warning squiggles and a Problems panel.
- Integrated Source Control: Git stage/unstage/commit/push/pull, branch switching and creation, an inline diff viewer, and AI-generated commit messages from the staged diff.
- Snippet and Emmet-style expansion across 7 languages; live code execution/preview (HTML, Markdown, CSS) with a sandboxed proxy for relative asset resolution.



Architecture & Infrastructure

- Flask backend with 50+ routes; 11-table SQLite schema: User (includes notes data), Chat, Memory, CalendarEvent, UploadedFile, ApiKey, Personality, Agent, Chain, EmailAlias, Conversation/Message; wiz_profile table with FK to User for structured profile data.
- CORS configured for wizardecosystem.dev and localhost; streaming via SSE; HTTP-only, SameSite=None, Secure cookies; rate limiting and graceful fallbacks.
- Electron Builder: NSIS installer (Windows), DMG (macOS), AppImage (Linux); GitHub Releases auto-updater; static frontends on GitHub Pages with custom dpdns.org DNS.

OTHER PROJECTS



A&A Web Studio — Static Portfolio Website

Jan 2026

- Designed and deployed a responsive static portfolio site at aaweb.neocities.org with structured navigation, JotForm contact integration, and a fully responsive layout.



Ultimate Clicker — Browser Incremental Game

Nov 2025

- Developed a browser-based incremental clicker game with score tracking, multipliers, level progression, and an upgrade shop (ultimate-clicker.w3spaces.com); dark-theme UI with animated progress bars.

SKILLS

Programming Languages: Python, JavaScript, HTML, CSS, Java

Frameworks & Tools: Flask, Electron, SQLAlchemy, Node.js, SQLite, GitHub & Version Control

AI & ML: Groq API (Llama 4 Scout), RAG, Prompt Engineering, Multi-Agent Orchestration, SSE Streaming

Platforms & Deployment: PythonAnywhere, GitHub Pages, Cloudflare Email Routing, Electron Builder, dpdns.org DNS

Mathematics: Competition-level; Algebra II, Trigonometry, Algorithmic Reasoning, Honors Precalculus

Typing: 45 WPM

Spoken Languages: English (fluent), Hindi (fluent), Japanese (beginner), Spanish (beginner)

Arts & Athletics: Violin, Piano, Recorder, Ukulele • Taekwondo (1st-Degree Black Belt) • Weightlifting, Gardening

EDUCATION

BASIS Phoenix

Grades 6 – Present

GPA: 4.0 · Distinguished Honour Roll (Top 5%)

- Grade 6: Algebra I, Geometry, Chemistry, Physics, Biology, Orchestra, Linguistics, World History
- Grade 7: Algebra II, Trigonometry, Chemistry, Physics, Biology, Linguistics, World History, Orchestra, Logic

BASIS Phoenix Primary

Grades 1 – 5

GPA: 4.0 · Distinguished Honour Roll (Top 5%) · Integrity Award · Grade 5: Pre-Algebra (High School Level)

EXTRACURRICULAR ACTIVITIES

Math Madness Competition Club · BASIS Phoenix

Aug 2026 – April 2027

- Participates in national math competitions and collaborative problem-solving.

Orchestra — Violin · BASIS Phoenix

Aug 2024 – Present

- Violin 1 / Violin 2 across Winter Concerts (2024, 2025), Festival of the Arts (2025, 2026), and Spring Concerts (2025, 2026).
- Repertoire includes: Momentum, Ukrainian Bell Carol, Beauty and the Beast, The Mandalorian, Strings of Fire, New Beginnings, Honour and Glory, Bradenburg Concerto 3(mvt 3)

ATA Martial Arts — Leadership & Discipline Program

June 2024 – Aug 2026

- First-Degree Black Belt (25 April 2026); Leadership Program and Life Skills Program.
- Specialties: Traditional & Creative Forms, Combat & Traditional Sparring, Bo Staff, Ssang Naht, Sword, Nunchucks, Weapons Forms.

Innovara Research Program · Science Fair & Research

Sept 2025 – Mar 2026

- Computer Science research track; Faraday's Law physics project.

Mathletics — Competition Preparation Club · BASIS Phoenix

Sept 2024 – Mar 2025

- Participates in MATHCOUNTS and MathLeague; refer to Competitions & Awards for placement details.

COMPETITIONS & AWARDS

Mathematics

- MathLeague: 4th place (Qualifier, December 2024); 9th place (District, February 2025); 11th place (State, May 2025).
- MATHCOUNTS Charter Competition: Top 20% of 200+ participants (February 2025).
- Purple Comet Math Meet: International Competition (March 2025).

Science & Technology

- BASIS Phoenix Science Fair (February 2026).
- Frostbyte Hackathon (February 2026).
- Code Arizona Hackathon (August 2026)

Taekwondo

- C Tournament (Yellow Belt): 1st Place Traditional Forms; 3rd Place Combat Sparring.

Performing Arts

- Singing Competition — 8th place out of 110 participants (local).

VOLUNTEER EXPERIENCE

Maha Ganpati Temple of Arizona • Maricopa, AZ

2023

- Maintained and sanitized community spaces, contributing to cleanliness and organization in service of the temple community.

Festival of the Arts • BASIS Phoenix

Feb 2025 & 2026

- Assisted with setup, seating, and equipment organization to ensure smooth and professional event operations.